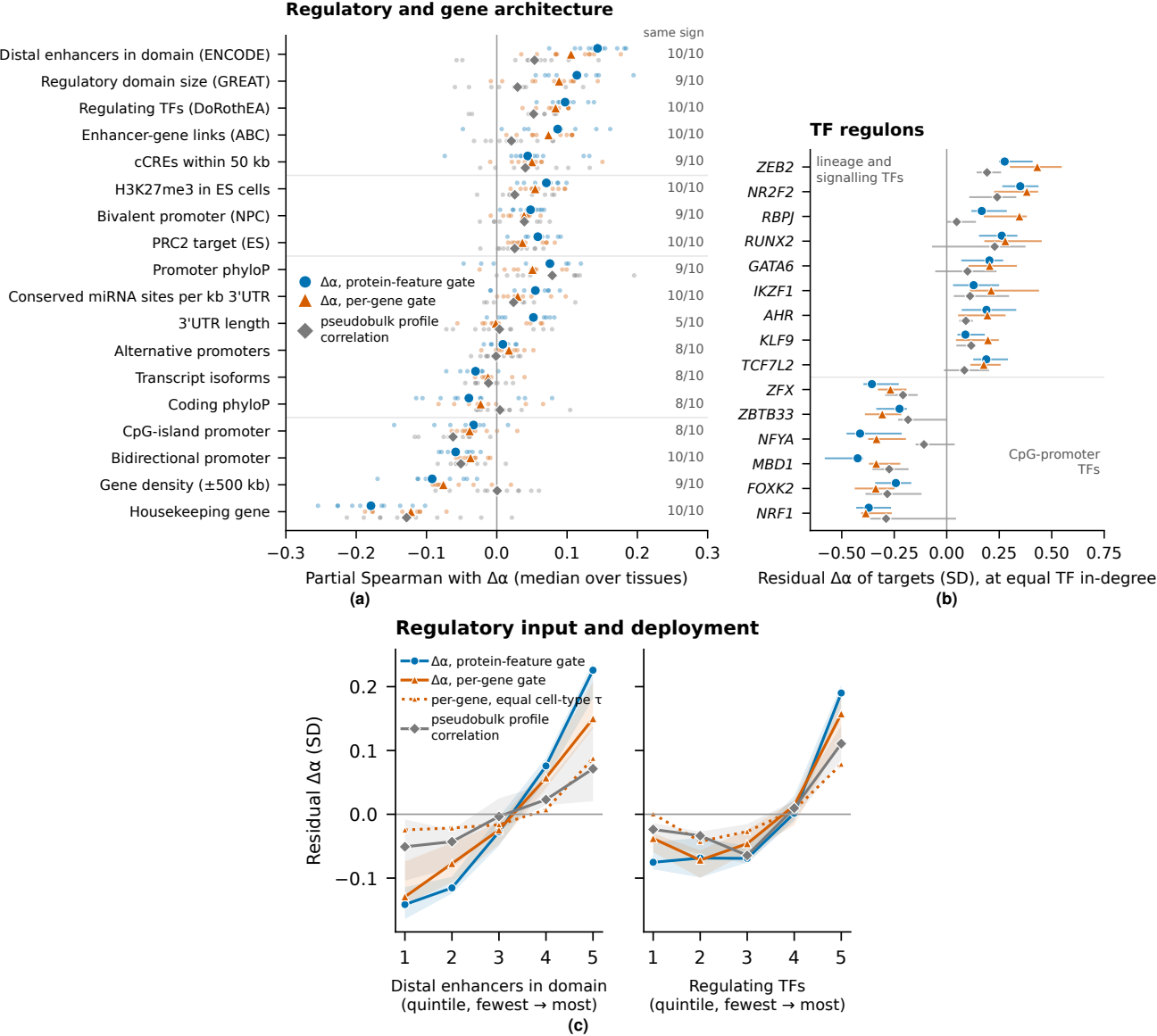

Regulatory and gene architecture
same sign
Distal enhancers in domain (ENCODE)
Regulatory domain size (GREAT)
Regulating TFs (DoRothEA)
Enhancer-gene links (ABC)
cCREs within 50 kb
H3K27me3 in ES cells
Bivalent promoter (NPC)
PRC2 target (ES)
Promoter phyloP
Conserved miRNA sites per kb 3'UTR
3'UTR length
Alternative promoters
Transcript isoforms
Coding phyloP
CpG-island promoter
Bidirectional promoter
Gene density (±500 kb)
Housekeeping gene
10/10
9/10
10/10
10/10
9/10
10/10
9/10
10/10
9/10
10/10
5/10
8/10
8/10
8/10
8/10
10/10
9/10
10/10
Δα, protein-feature gate
Δα, per-gene gate
pseudobulk profile correlation
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Partial Spearman with Δα (median over tissues)
(a)
TF regulons
lineage and signalling TFs
ZEB2
NR2F2
RBPJ
RUNX2
GATA6
IKZF1
AHR
KLF9
TCF7L2
ZFX
ZBTB33
NFYA
MBD1
FOXK2
NRF1
CpG-promoter TFs
−0.50
−0.25
0.00
0.25
0.50
0.75
Residual Δα of targets (SD), at equal TF in-degree
(b)
Regulatory input and deployment
Δα, protein-feature gate
Δα, per-gene gate
per-gene, equal cell-type τ
pseudobulk profile correlation
Residual Δα (SD)
0.2
0.1
0.0
−0.1
1
2
3
4
5
Distal enhancers in domain (quintile, fewest → most)
Regulating TFs (quintile, fewest → most)
(c)